

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Login	Allows applicants and officers to securely register and log in using email/password with JWT authentication.	Authentication Module	Done	1
2	Loan Application Form	Applicants can submit personal, financial, and loan details via a web form.	Frontend (Flask UI)	Done	1
3	Data Preprocessing	Cleans and validates applicant data before feeding into ML models.	ML Pipeline	Done	1
4	ML Prediction Engine	Predicts applicant credibility using trained models (XGBoost as final).	Backend ML Service	Done	1
5	Real-Time Decision Dashboard	Displays prediction results to credit officers for faster approvals.	Officer Dashboard	Done	1
6	Reporting & Analytics	Generates reports on approval rates, risk scores, and model performance.	Reporting Module	In Progress	0.5

7	Audit Logging	Stores prediction logs for compliance and monitoring.	Database / Logging Service	Done	0.5
8	External API Integration	Optional integration with credit history APIs for enhanced profiling.	API Layer	Pending	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6 (core features complete)
Core / Must-Have Features	5 (registration, login, loan form, ML prediction, dashboard)
Additional / Nice-to-Have Features	3 (reporting, audit logging, external API integration)
Features Tested & Verified	6 (all core features tested and functional)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Registration/Login form, Loan Application form
2	Backend / Logic	3	Data Preprocessing, ML Prediction Engine, Decision Dashboard
3	Database / Storage	2	Applicant Database, Prediction Logs
4	API / Integration	1	External Credit History API
5	Security / Authentication	1	JWT-based secure login